

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Embedded Systems
COURSE CODE: ECA1422

COURSE OUTCOME COVERED

CO1: *Analyze and design complex and networked embedded systems using formal design methodologies, embedded architectures, CAN (BL5)*

Total Marks: 20

Sl. No	QUESTION BANK	BLOOMS LEVEL
	Create a Concept map that models the hardware-software integration of system validation phase of the embedded product development cycle - connection integration, strategies, test benches, simulation environments, hardware-in-loop setups, acceptance criteria	

Code of Conduct:

M. Jahnvi / 192411196 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Jahnvi

Evaluation Rubric (20 Marks per Question – Concept Map)

Criteria	Weightage	Level 4 (Exemplary) (5 marks)	Level 3 (Proficient) (4 marks)	Level 2 (Developing) (3 marks)	Level 1 (Beginning) (1-2 marks)
Concept Identification	25 %	20			
Linkages	25 %	20			
Organization	20 %	20			
Integration of Literature	20 %	20			
Clarity	10%	10			

100



HARDWARE-SOFTWARE INTEGRATION AND SYSTEM VALIDATION IN EMBEDDED PRODUCT DEVELOPMENT - CONCEPT MAP

Modeling Integration Strategies, Test Environments, Validation, Compliance & Product Release Gates

- HOW EACH PHASE CONTRIBUTES
- ★ Functional Correctness
 - ◎ Safety Assurance
 - Ⓢ Regulatory Compliance
 - ✓ Reliable Product Release

1. SYSTEM INTEGRATION

- Hardware Integration
- Software Integration
- Interface Integration
- Synchronization
- Data & Control Flow

Seamless interaction between hardware, software and interfaces

2. INTEGRATION STRATEGIES

- Incremental Integration
- Top-Down Integration
- Bottom-Up Integration
- Sandwich Integration
- Continuous Integration (CI/CD)

Reduces integration risk and early detection of defects

3. SIMULATION ENVIRONMENTS

- Software Simulation
- MATLAB/Simulink
- Proteus / Virtual Labs
- QEMU / Virtual MCU
- Digital Twin Models

Validate behavior before hardware availability and reduce cost

4. TEST BENCHES

- Signal Generator
- Oscilloscope
- Logic Analyzer
- Power Supply
- Data Logger
- Automated Scripts

Controlled setup to test hardware & software modules

9. PRODUCT RELEASE GATE

- Design Reviews
- Integration Complete
- Validation Complete
- Safety Certification
- Regulatory Approval
- Production Readiness
- Customer Acceptance
- Market Release

Final approval to release safe, reliable and compliant embedded product

8. REGULATORY STANDARDS & COMPLIANCE

- ISO 26262 (Automotive Safety)
- IEC 62304 (Medical Devices)
- IEC 61508 (Functional Safety)
- MISRA C (Coding Standards)
- EMC / EMI Standards
- CE / FCC Certification
- Cybersecurity Standards

Ensures mandatory safety, quality and regulatory compliance

INTEGRATION BENEFITS

- ★ Reduced Errors
- ★ Faster Development
- ★ Reliable Communication
- ★ Early Defect Detection
- ★ Improved Quality

FORMAL VALIDATION (ENSURES)

- Functional Correctness
- System Reliability
- Safety Assurance
- Risk Reduction
- Regulatory Compliance
- Product Certification

VALIDATION BENEFITS

- ★ Meets ISO 26262
- ★ Meets IEC 62304
- ★ Improved Product Quality
- ★ Customer Satisfaction
- ★ Safe Product Release

7. ACCEPTANCE CRITERIA

- All Functional Requirements Met
- No Critical / Major Defects
- Performance Targets Achieved
- Timing Constraints Met
- Power Budget Satisfied
- Safety Goals Achieved
- Reliability Targets Achieved
- Regulatory Standards Met
- User Acceptance Completed
- Documentation Complete
- Production Readiness Confirmed

Confirms that the system meets all critical safety & quality requirements

6. SYSTEM VALIDATION

- Functional Testing
- Integration Testing
- Regression Testing
- Stress & Performance Testing
- Security Testing
- EMI/EMC Testing
- Environmental Testing
- Fault Tolerance Testing
- Reliability Testing
- Power Consumption Testing

Verifies that the integrated system works as intended under all conditions

5. HARDWARE-IN-LOOP (HIL) SETUPS

- Real Controller
- Simulated Environment
- Sensors / Actuators
- Real-time Interface
- Fault Injection

Real-time testing with actual hardware in the loop

TOOLS & SUPPORT

- Model-Based Design (MBD)
- Automated Testing
- Continuous Integration
- Version Control (Git)
- Static Code Analysis
- Requirements Management

RESULT

A safe, reliable, high-quality and standards-compliant embedded product ready for deployment and market success.

HOW FORMAL VALIDATION ENSURES SAFETY & CORRECTNESS

- ✓ Verifies every requirement → Functional Correctness
- ✓ Identifies hazards and mitigates risks → Safety Assurance
- ✓ Ensures compliance with standards → Regulatory Compliance
- ✓ Validates performance in real-world conditions → Reliability
- ✓ Provides evidence for certification → Product Certification
- ✓ Builds confidence for customers and stakeholders

FORMAL VALIDATION GUARANTEES FUNCTIONAL CORRECTNESS, SAFETY AND RELIABILITY OF THE FINAL EMBEDDED PRODUCT.



Following this integration and validation process ensures a safe, reliable, compliant and market-ready embedded product.

Good